Independent Study Course (NST3901)

Theoretical Foundations for 3D Reconstruction of Sports Games with Limited Visual Data Settings

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Nguyen Hoang The Kiet

NUS College, National University of Singapore

Contents

[1]Adopted from the ISC application form

1. Introduction

1.1. Motivation[2]

The Video Assistant Referee (VAR) system was introduced at the 2018 FIFA World Cup in Russia, after which it became popularised in other European domestic and continental leagues. Originally a video review system for referees in case of critical fouls (e.g., offsides leading to a goal, offences in the penalty box, etc.), the technology has been equipped with major upgrades, such as the semi-automated offside technology (SAOT) used in the latest edition of the FIFA World Cup in Qatar [1].

Figure 1: Semi-automated offside technology used in Italy's Serie A.

Lower-tier tournaments such as the 2024 ASEAN Cup (formerly known as AFF Cup) and domestic leagues in Southeast Asia are also improving transparency in refereeing decisions by implementing low-budget versions of the standard Video Assistant Referee (VAR), such as VAR Lite or centralised VAR. However, these budget systems quickly showed vulnerability in operation, especially in high-stakes situations. For instance, in the two legs of the Thailand vs. Philippines match-off in the 2nd semifinal of the ASEAN Cup:

- In the first leg [2], VAR did not participate in a potential offside offence by the Filipino player Alex Munis, which then led to the Philippines' first goal in their historic 2-1 win against Thailand. Speculations from social media then showed that VAR seemed not to have worked at all due to the latency between the field and the centralised VAR station in Singapore.
- In the second leg [3], VAR also did not intervene when the Thai player Seksan Ratree allegedly dribbled the ball crossing the goal line, leading to Peradol's opening goal for Thailand that would win them the leg and the semi-final. The replay was indecisive and caused many controversies afterwards.

Figure 2: Two of the controversial incidents in the Thailand v Philippines ASEAN Cup semi-final legs.

In light of the mentioned shortcomings, this project aims to build an end-to-end solution for reconstructing game events in 3D under limited data conditions, with potential applications in refereeing, analytics and broadcasting.

1.2. Scope of the Independent Study Course project

The first stage of the project will be a survey of foundational methods in the field registration stage. The aim is to reconstruct a model of a football field

[2] Adopted from the ISC application form

1.3. Dataset & Problem analysis

1.3.1. Dataset

1.3.2. Suggested pipeline

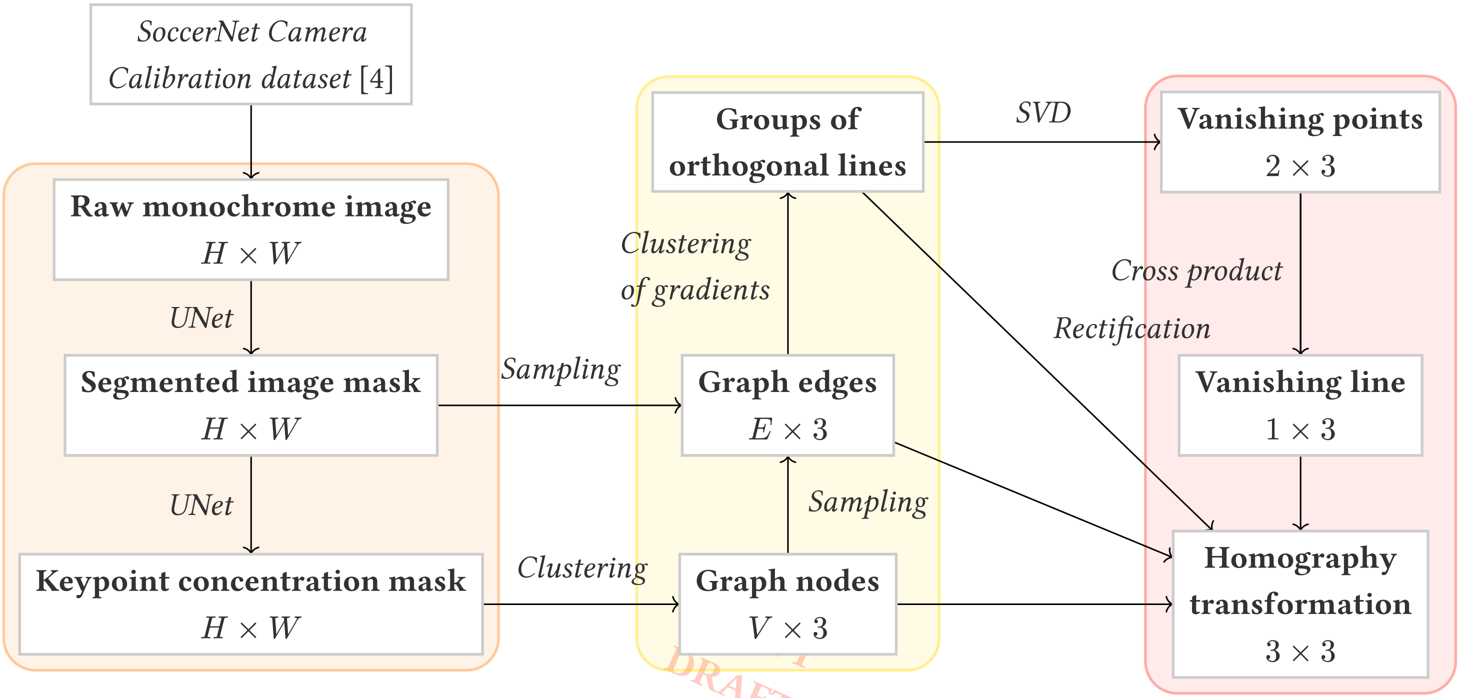


Figure 3: Summary of the suggested automated pipeline for field registration

We propose the above automated pipeline for the field registration problem, which can be divided into three smaller subtasks with its respective domain of knowledge:

- **Feature masks generation**. [5]
- **Graph creation**. [6]
- **Homography reconstruction**. [7]

1.3.3. Limitations and Mitigations

During the course of implementation,

- ..

2. Feature masks generation via generative models

3. Detecting keypoints and edges of the skeleton graph via clustering

After segmentation and keypoint concentration, we obtain two image masks: an outline mask and a keypoint concentrated mask. ---

3.1. Detecting keypoints through clustering methods

From the keypoint concentrated mask, we extract all coordinates $\left(x^{(i)}, y^{(i)}\right)$ whose pixel intensity level exceeds a threshold level T ($T \approx 0.8$). We shall thus obtain a list of "white" pixels $\boldsymbol{P} = [\vec{\boldsymbol{p}}_1 \ \vec{\boldsymbol{p}}_2 \ \cdots \ \vec{\boldsymbol{p}}_m]$. This list is yet the desired keypoints list, instead we observe from the mask that each keypoint is associated with a "cluster" of "white" pixels, which means that a keypoint is the average of some "white" points $\vec{\boldsymbol{p}}_i$ within its clutser.

The above observation motivates us for a keypoint positioning algorithm via clustering methods as follows:

Input.

- Keypoint concentrated mask I_K of size $H \times W$.
- Threshold $T \approx 0.8$

Output. List of keypoints $P = \begin{bmatrix} x^{(1)} & x^{(2)} & \dots & x^{(n)} \\ y^{(1)} & y^{(2)} & \dots & y^{(n)} \end{bmatrix}$.

Hyperparameters. A clustering algorithm $\mathcal{C}$.

Step 1. From the keypoint concentrated mask, extract all coordinates $\left(x^{(i)}, y^{(i)}\right)$ whose pixel intensity level exceeds a threshold level T, thus obtaining a list of "white" pixels $\boldsymbol{Q} = \left[\vec{\boldsymbol{q}}^{(1)} \ \vec{\boldsymbol{q}}^{(2)} \ \dots \ \vec{\boldsymbol{q}}^{(m)}\right]$.

Step 2. Run the clustering algorithm $\mathcal{C}$ on $\boldsymbol{Q}$, extract all **cluster centroids** $\boldsymbol{P} = \left[\vec{\boldsymbol{p}}^{(1)} \ \vec{\boldsymbol{p}}^{(2)} \ \dots \ \vec{\boldsymbol{p}}^{(m)}\right]$.

A similar approach can be seen in [6].

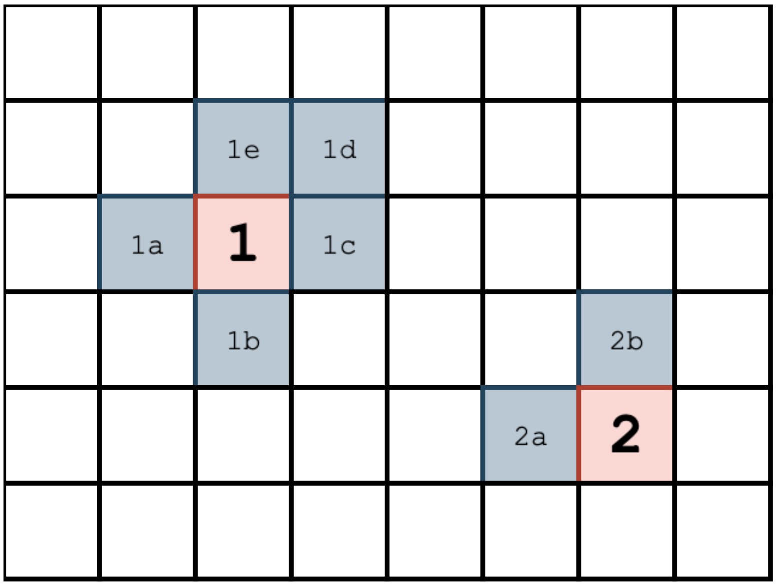


Figure 4: Clustering method for keypoint detection

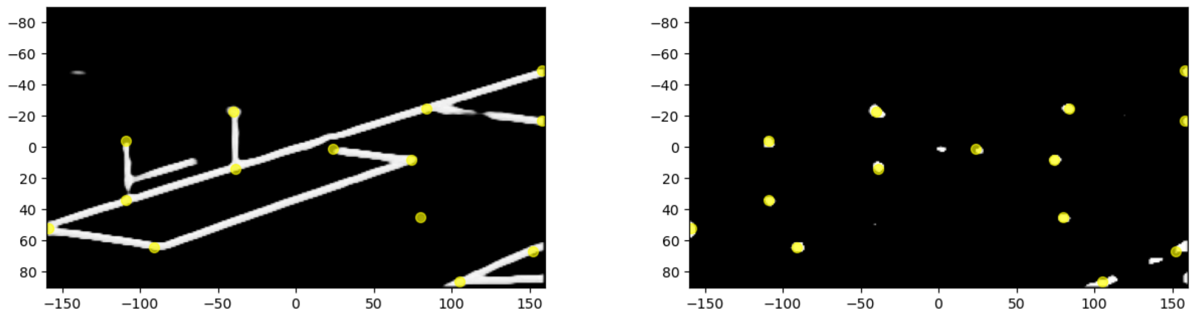


Figure 5: Outline mask (left) and keypoint concentrated mask (right)

3.2. Detecting edges through sampling

For each pair of keypoints, we want to establish a graph of connecting edges between the keypoints if there is a line segment connecting them on the image. As the segmented mask has seperated the filed lines with the rest of the image, yet still noisy, we need a robust algorithm to detect an edge between any keypoints pair.

Our idea is to sample as many points as possible lying between the two keypoints $\vec{p}_1$ and $\vec{p}_2$. Such sampled points will take the form $\vec{q} = (1 - t)\vec{p}_1 + t\vec{p}_2$, where t is a parameter within the $[0, 1]$ range. For additional robustness, an error term $\vec{\varepsilon}$ is added to the sampled point $\vec{q}$. This prevents us from misdetecting edges whose endpoints lie at the border of the edge (as all edges in this case have thickness, therefore it matters where the keypoints are located). The number of sampled points, T, can be taken in proportion with the length of the segment $\vec{p}_1\vec{p}_2$.

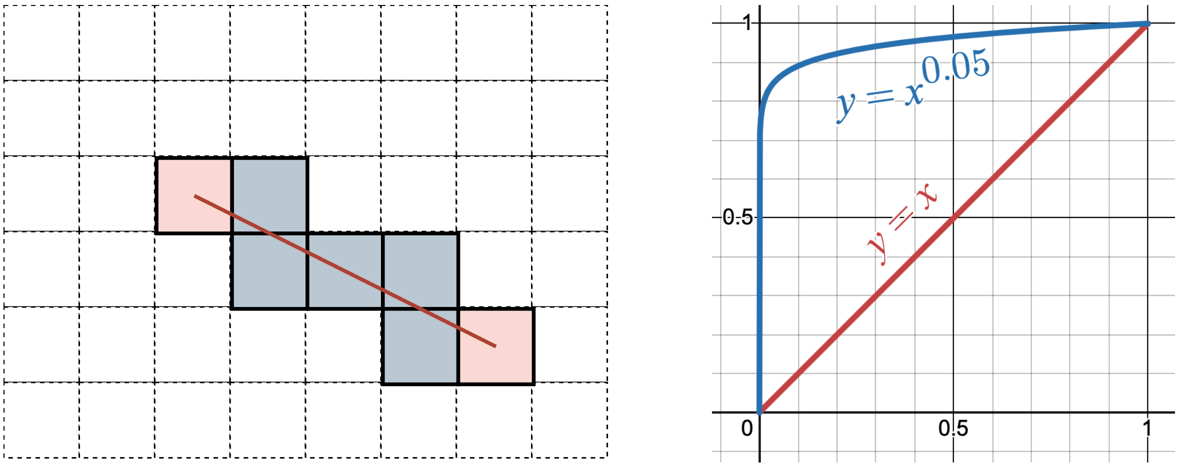


Figure 6: Clustering method for keypoint detection

Input.

- Edge mask I_E of size $H \times W$;
- Two keypoints $\vec{\boldsymbol{p}}_1$ and $\vec{\boldsymbol{p}}_2$;
- $\gamma \approx 0.01$

Output. Confidence if there is an edge connecting $\vec{\boldsymbol{p}}_1$ and $\vec{\boldsymbol{p}}_2$, between 0 and 1.

Step 1. Let $T := \lceil \|\vec{\boldsymbol{p}}_1 - \vec{\boldsymbol{p}}_2\| \rceil$.

Step 2. Generate T random values of $t^{(i)} \in [0, 1]$.

Step 3. For each $t^{(i)}$, the sampled point is $\vec{\boldsymbol{q}}^{(i)} := \left(1 - t^{(i)}\right)\vec{\boldsymbol{p}}_1 + t^{(i)}\vec{\boldsymbol{p}}_2 + \vec{\varepsilon}$, where $\vec{\varepsilon}$ is a random noise vector, taking range $[-2, 2]$ for each dimension.

Step 4. Calculate the sum of intensities of the sampled points, up to a power of γ:

$$\text{Confidence} := \sum_i I_E\left(q_x^{(i)}, q_y^{(i)}\right)^\gamma$$

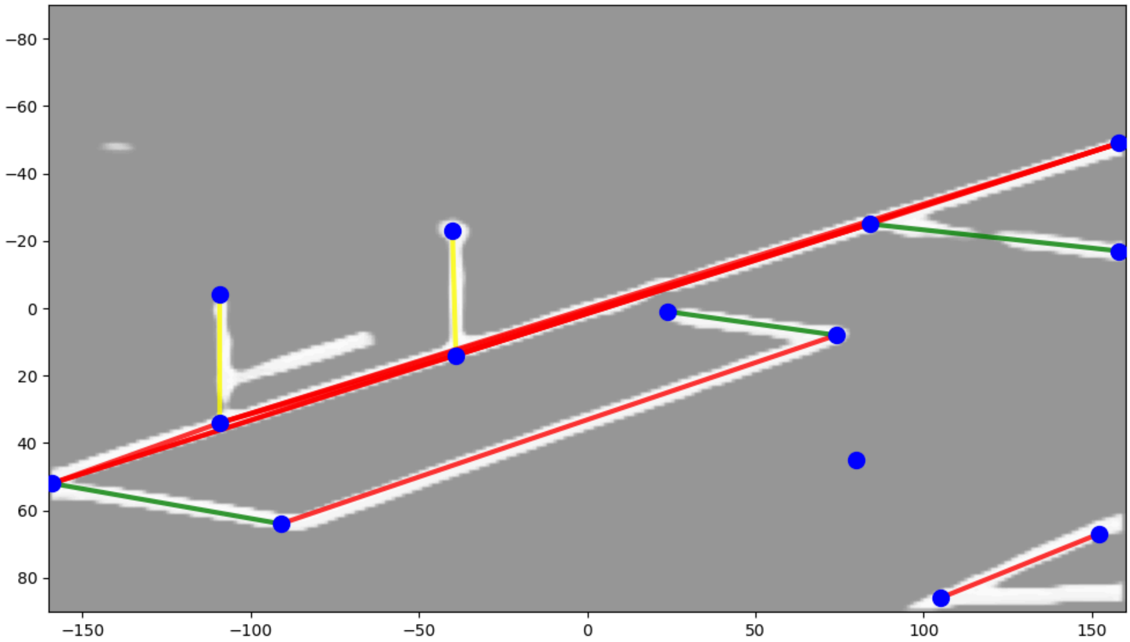


Figure 7: Classified segments based on the gradient vector
(red for vertical field lines, green for horizontal field lines, and yellow for upright)

3.3. Detecting groups of orthogonal lines via clustering of gradients

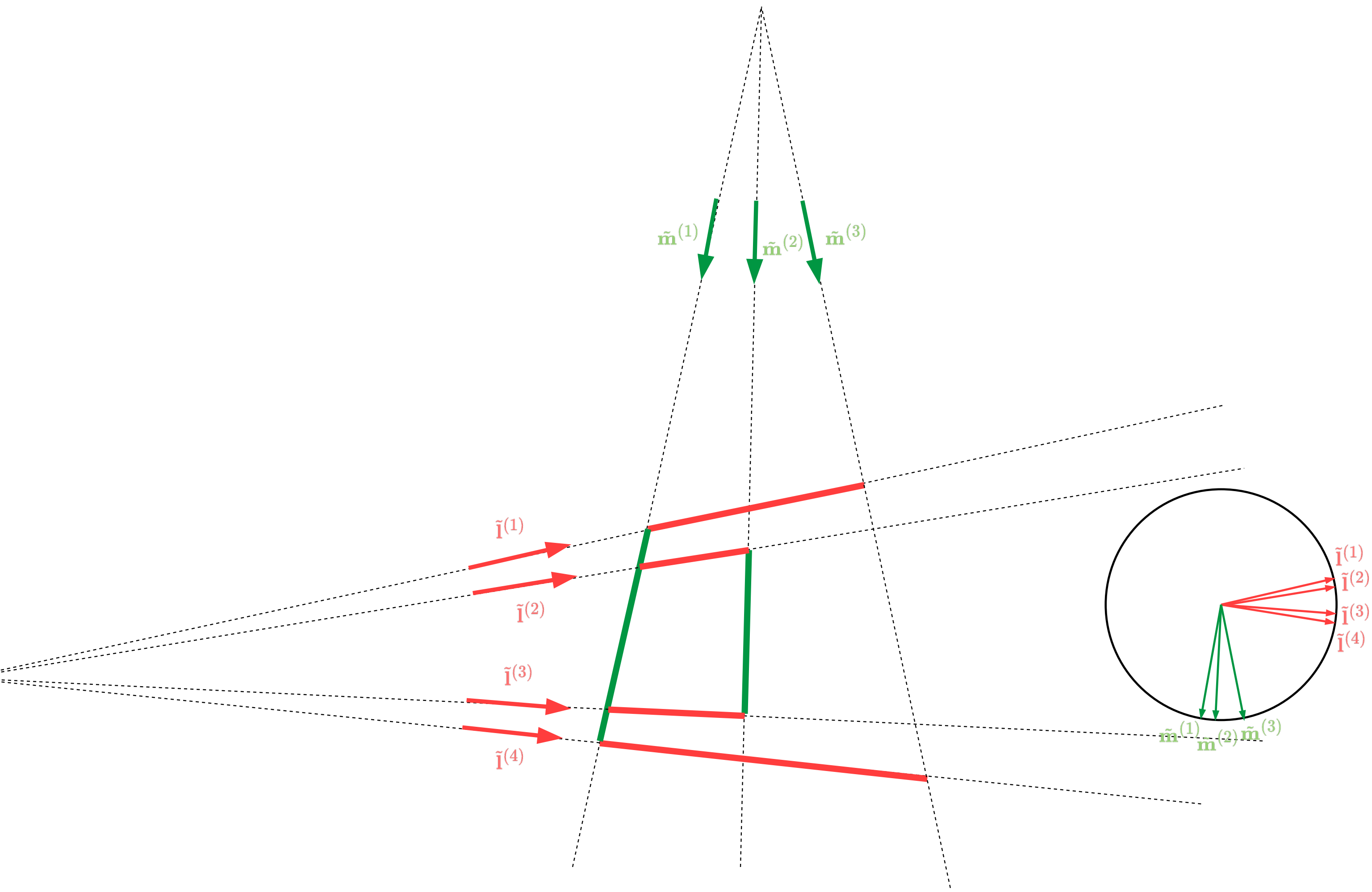


Figure 8: Classified segments based on the gradient vector
(red for vertical field lines, green for horizontal field lines, and yellow for upright)

Grouping lines via gradients

Input.

- Edge mask I_E of size $H \times W$;
- Two keypoints $\vec{\boldsymbol{p}}_1$ and $\vec{\boldsymbol{p}}_2$;
- $T \approx 1000$;
- $\gamma \approx 0.01$

Output. Confidence if there is an edge connecting $\vec{\boldsymbol{p}}_1$ and $\vec{\boldsymbol{p}}_2$, between 0 and 1.

Step 1. Generate T random values of $t^{(i)} \in [0, 1]$.

Step 2. For each $t^{(i)}$, the sampled point is $\vec{\boldsymbol{q}}^{(i)} := \left(1 - t^{(i)}\right)\vec{\boldsymbol{p}}_1 + t^{(i)}\vec{\boldsymbol{p}}_2$.

Step 3. Calculate the sum of intensities of the sampled points, up to a power of γ:

$$\text{Confidence} := \sum_i I_E\left(q_x^{(i)}, q_y^{(i)}\right)^\gamma$$

Step 4. Return the confidence.

4. Homographic rectification

For theoretical and computational methods regarding the homographic transformation, the report refers to Hartley and Zisserman [7, Chapter 2] on *Projective geometry and Transformations of 2D*.

4.1. Projective geometry

Consider the $\mathbb{R}^3$ space with the plane $(\pi) : z = 1$, which we denote as the **projective plane**. Let $\mathcal{H}$ be a set of points in $\mathbb{R}^3$ (a single point, a line, a shape, etc.) in $\mathbb{R}^3$. Then, for each point $\tilde{\boldsymbol{p}}$ in $\mathcal{H}$, the line through $\tilde{\boldsymbol{p}}$ and the origin intersects the projective plane (π) at a point denoted $\vec{\boldsymbol{p}}$. *We call $\vec{\boldsymbol{p}}$ the projective image of $\tilde{\boldsymbol{p}}$ onto the projective plane.*

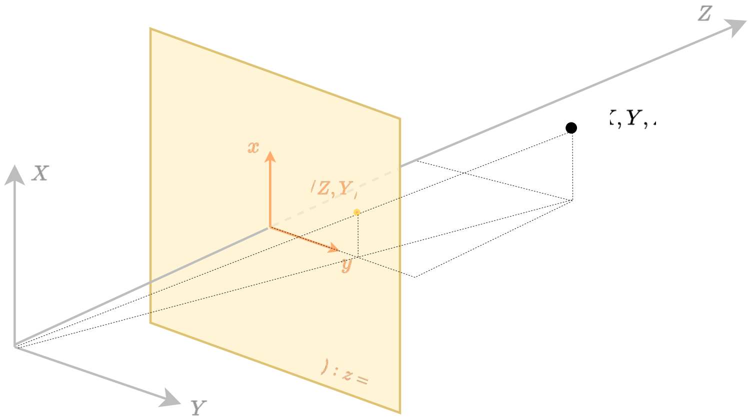


Figure 9:

This projection motivates us to derive an alternative representation for any point $\vec{\boldsymbol{p}} = [x \;\; y]^T$ on a 2-dimensional (2D) plane by considering it the projective image onto the plane $z = 1$ of any point $\tilde{\boldsymbol{p}}$ in $\mathbb{R}^3$. The coordinates of point $\vec{\boldsymbol{p}}$ in $\mathbb{R}^3$ is $[x \;\; y \;\; 1]^T$ as $\vec{\boldsymbol{p}}$ lies on the $z = 1$ plane. Using the argument of similar triangles, one can derive that all possible representations of the correspondent finite point $\tilde{\boldsymbol{p}}$ is thus $k[x \;\; y \;\; 1]^T = [kx \;\; ky \;\; k]^T$ for any $k \neq 0$.

Following this representation, any line $ax + by + c = 0$ on the $\mathbb{R}^2$ plane can also be uniquely defined by a single homogenous vector $\tilde{\boldsymbol{l}} = [a \;\; b \;\; c]^T$. This gives us the following important, yet computationally neat, corollaries:

- A point $\tilde{\boldsymbol{p}}$ is incident with (or lies on) a line $\tilde{\boldsymbol{l}}$ iff $\tilde{\boldsymbol{l}}^T \tilde{\boldsymbol{p}} = 0$.
- Two lines $\tilde{\boldsymbol{l}}_1$ and $\tilde{\boldsymbol{l}}_2$ always intersects at point $\tilde{\boldsymbol{p}} = \tilde{\boldsymbol{l}}_1 \times \tilde{\boldsymbol{l}}_2$ (whether the intersection actually corresponds to a real point on the 2D plane is a different problem).

- The line crossing two points $\tilde{\boldsymbol{p}}_1$ and $\tilde{\boldsymbol{p}}_2$ is computed as $\tilde{\boldsymbol{l}} = \tilde{\boldsymbol{p}}_1 \times \tilde{\boldsymbol{p}}_2$.

Note that the second corollary also works for parallel lines, that is, the two lines of form $\tilde{\boldsymbol{l}}_1 = \begin{bmatrix} a & b & c_1 \end{bmatrix}^T$ and $\tilde{\boldsymbol{l}}_2 = \begin{bmatrix} a & b & c_2 \end{bmatrix}^T$. The intersection of which is,

$$\tilde{\boldsymbol{l}}_1 \times \tilde{\boldsymbol{l}}_2 = (c_2 - c_1) \begin{bmatrix} b \\ a \\ 0 \end{bmatrix}$$

Thus, a homogenous vector whose last component is zero can be thought of as the "intersection" of parralel lines, representing "the point at infinity."

The projective space is also denoted as $\mathbb{P}^2 = \mathbb{R}^3 - \{(0,0,0)\}$.

4.2. Transformations in 2D

A geometric transformation T is a mapping from a shape $\mathcal{H}$ to its image $\mathcal{H}'$, preserving certain geometric identities between the original and the transformed shape.

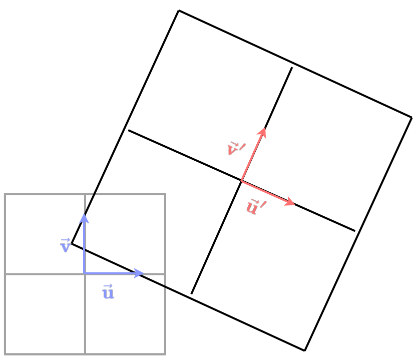
We consider linear transformations in the homogenous coordinates system $\mathbb{P}^2$. These transformations take the form

$$\begin{aligned} H : \mathbb{P}^2 &\mapsto \mathbb{P}^2 \\ \tilde{\boldsymbol{p}} &\mapsto H\tilde{\boldsymbol{p}} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \end{aligned}$$

where H

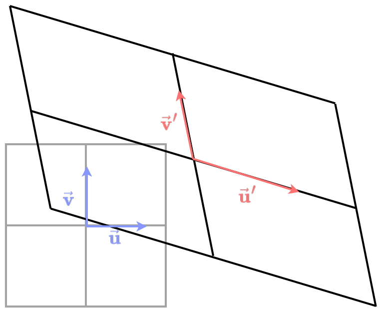
We consider the following 3 families of transformation:

- **Similarity**. A similarity is a transformation that preserves angles and length ratios. It can be further decomposed as a chain of a θ-rotation about the origin $R(\theta)$, followed by a scale k and a translation along the vector $\vec{\boldsymbol{v}}$.



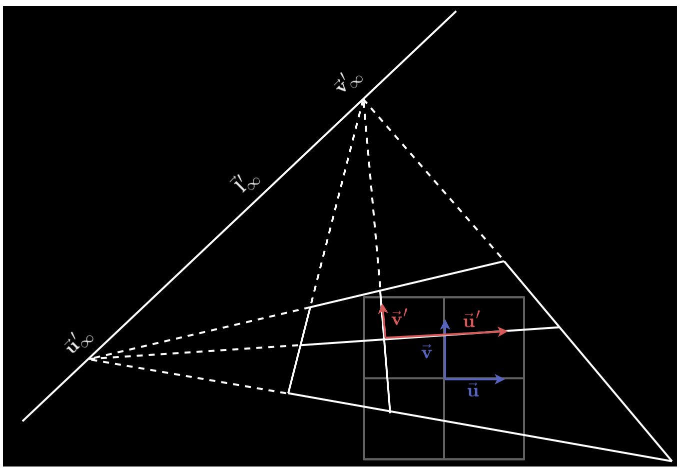
$$\tilde{\boldsymbol{p}} = \begin{bmatrix} k\cos\theta & -k\sin\theta & v_x \\ k\sin\theta & k\cos\theta & v_y \\ 0 & 0 & 1 \end{bmatrix} \tilde{\boldsymbol{p}}$$

- **Affinity**. An affinity is a transformation that preserves parallelisms and length ratios. It can be described as a linear transformation in $\mathbb{R}^2$ where the basis $\{\hat{\imath}, \hat{\jmath}\}$ is transformed into $\{a_{11}\hat{\imath} + a_{21}\hat{\jmath}, a_{12}\hat{\imath} + a_{22}\hat{\jmath}\}$. A similarity is also a special case of an affinity where $a_{11} = a_{22} = k\cos\theta$ and $-a_{12} = a_{22} = k\sin\theta$.



$$\tilde{\boldsymbol{p}} = \begin{bmatrix} a_{11} & a_{12} & v_x \\ a_{21} & a_{22} & v_y \\ 0 & 0 & 1 \end{bmatrix} \tilde{\boldsymbol{p}}$$

- **Homography**. A homography is a transformation that preserves incidence and collinearity, that is, if a point $\vec{\boldsymbol{p}}$ belongs to the line $\tilde{\boldsymbol{l}}$ (or $\tilde{\boldsymbol{p}}^T\tilde{\boldsymbol{l}} = 0$), then so is $\vec{\boldsymbol{p}}'$ on the transformed line $\tilde{\boldsymbol{l}}'$ An affinity is a special case of a homography where $h_{31} = h_{32} = 0$.



$$\tilde{\boldsymbol{p}} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \tilde{\boldsymbol{p}}$$

In field registraion we are interested in recovering the homography matrix in two conditions:

- Condition 1 (DLT)
- Condition 2 (Rectification + Bruteforce RANSAC)

4.3. Solving linear systems using the Singular Value Decomposition (SVD)

Often when solving for parameters of a certain geometric entity or transformation, it is needed to solve for the non-trivial roots of the correspondent linear system, for instance, $A\vec{\boldsymbol{x}} = \vec{\boldsymbol{0}}$. However, due to noise and errors arising from preceding calculations, the system can either be underdetermined or overdetermined.

A common workaround is to frame the problem as a minimisation of the norm of the residual vector $\|A\hat{\boldsymbol{x}}\|^2$, with the restriction $\|\hat{\boldsymbol{x}}\|^2 = 1$ to avoid the trivial solution $\hat{\boldsymbol{x}} = \boldsymbol{0}$. One common method for these types of problem is the Singular Value Decomposition (SVD) algorithm.

From the SVD, any matrix $A_{m \times n}$ can be written as

$$A = UDV^T = \sum_{i=1}^{n} \sigma_i \hat{\boldsymbol{u}}_i \hat{\boldsymbol{v}}_i^T$$

where $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ are the singular values of A, and $U = [\hat{\boldsymbol{u}}_1 \;\; \hat{\boldsymbol{u}}_2 \;\; \cdots \;\; \hat{\boldsymbol{u}}_r]$, and $V = [\hat{\boldsymbol{v}}_1 \;\; \hat{\boldsymbol{v}}_2 \;\; \cdots \;\; \hat{\boldsymbol{v}}_r]$ satisfying $U^T U = V^T V = I$.

This representation gives us a straightforward solution: $\arg \min_{\hat{\boldsymbol{x}}:\, \|\hat{\boldsymbol{x}}\|=1} \|A\hat{\boldsymbol{x}}\|^2 = \hat{\boldsymbol{v}}_r$, i.e. the right vector associated with the least non-zero singular value. A short proof is given in [7, A5.3].

4.4. Recovering homography using point-to-point correspondence (Direct Linear Transform, DLT)

Given k point-to-point correspondences $\vec{\boldsymbol{p}}^{(i)} \leftrightarrow \vec{\boldsymbol{q}}^{(i)}$, where $k \geq 4$, find a homographic transformation H that maps every point $\vec{\boldsymbol{p}}^{(i)}$ to $\vec{\boldsymbol{q}}^{(i)}$.

Note that each correspondence $\vec{\boldsymbol{p}}^{(i)} \leftrightarrow \vec{\boldsymbol{q}}^{(i)}$ provides the equation $H\tilde{\boldsymbol{p}}^{(i)} \sim \tilde{\boldsymbol{q}}^{(i)}$ or

$$H\tilde{\boldsymbol{p}}^{(i)} \times \tilde{\boldsymbol{q}}^{(i)} = \tilde{\boldsymbol{0}}$$

$$\Leftrightarrow \begin{cases} \left(h_{11}p_1^{(i)} + h_{12}p_2^{(i)} + h_{13}p_3^{(i)}\right) q_2^{(i)} - \left(h_{21}p_1^{(i)} + h_{22}p_2^{(i)} + h_{23}p_3^{(i)}\right) q_1^{(i)} = 0 \\ \left(h_{21}p_1^{(i)} + h_{22}p_2^{(i)} + h_{23}p_3^{(i)}\right) q_3^{(i)} - \left(h_{31}p_1^{(i)} + h_{32}p_2^{(i)} + h_{33}p_3^{(i)}\right) q_2^{(i)} = 0 \\ \left(h_{31}p_1^{(i)} + h_{32}p_2^{(i)} + h_{33}p_3^{(i)}\right) q_1^{(i)} - \left(h_{11}p_1^{(i)} + h_{12}p_2^{(i)} + h_{13}p_3^{(i)}\right) q_3^{(i)} = 0 \end{cases}$$

$$\Leftrightarrow \begin{bmatrix} p_1^{(i)}q_2^{(i)} & p_2^{(i)}q_2^{(i)} & p_3^{(i)}q_2^{(i)} & -p_1^{(i)}q_1^{(i)} & -p_2^{(i)}q_1^{(i)} & -p_3^{(i)}q_1^{(i)} & & & \\ & & & p_1^{(i)}q_3^{(i)} & p_2^{(i)}q_3^{(i)} & p_3^{(i)}q_3^{(i)} & -p_1^{(i)}q_2^{(i)} & -p_2^{(i)}q_2^{(i)} & -p_3^{(i)}q_2^{(i)} \\ -p_1^{(i)}q_3^{(i)} & -p_2^{(i)}q_3^{(i)} & -p_3^{(i)}q_3^{(i)} & & & & p_1^{(i)}q_1^{(i)} & p_2^{(i)}q_1^{(i)} & p_3^{(i)}q_1^{(i)} \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ \vdots \\ h_{31} \\ h_{32} \\ h_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Leftrightarrow A^{(i)}\tilde{\boldsymbol{h}} = \tilde{\boldsymbol{0}}$$

Note that the left matrix $A^{(i)}$ has rank 2, therefore we only need to keep two of the three rows. As a homographic transformation H has 8 DoF, a minimum of 4 correspondences is required for the system to be determined.

The Direct Linear Transform (DLT) algorithm.

Input. k correspondences $p^{(i)} \mapsto q^{(i)}$ under the homographic transformation H.
Output. Best-fitting homography matrix H.

Step 1. For each correspondence $\tilde{\boldsymbol{p}}^{(i)} \leftrightarrow \tilde{\boldsymbol{q}}^{(i)}$, prepare the matrix

$$A^{(i)} = \begin{bmatrix} -p_1^{(i)} q_3^{(i)} & -p_2^{(i)} q_3^{(i)} & -p_3^{(i)} q_3^{(i)} & & & & p_1^{(i)} q_1^{(i)} & p_2^{(i)} q_1^{(i)} & p_3^{(i)} q_1^{(i)} \\ & & & p_1^{(i)} q_3^{(i)} & p_2^{(i)} q_3^{(i)} & p_3^{(i)} q_3^{(i)} & -p_1^{(i)} q_2^{(i)} & -p_2^{(i)} q_2^{(i)} & -p_3^{(i)} q_2^{(i)} \end{bmatrix}$$

Step 2. Stack all the matrices $A^{(i)}$ into a single matrix A of size $2k \times 9$:

$$A = \begin{bmatrix} A^{(1)} \\ A^{(2)} \\ \vdots \\ A^{(k)} \end{bmatrix}$$

Step 3. Solve for $\tilde{\boldsymbol{h}} = \arg\min_{\|\tilde{\boldsymbol{h}}'\|=1} \|A\tilde{\boldsymbol{h}}'\|^2$ via SVD. *If $k = 4$, the solution is exact.*

Step 4. Reshape $\tilde{\boldsymbol{h}}$ from size 9×1 to the matrix $H \in \mathbb{R}^{3\times 3}$.

4.5. Rectification

Given two groups of parallel lines $L = [\cdots\ \tilde{\boldsymbol{l}}^{(i)}\ \cdots]$ and $M = [\cdots\ \tilde{\boldsymbol{m}}^{(j)}\ \cdots]$ such that each pair of lines $\left(\tilde{\boldsymbol{l}}^{(i)}, \tilde{\boldsymbol{m}}^{(j)}\right)$ is orthogonal. Under the homography H, the groups become $L' = HL = [\cdots\ \tilde{\boldsymbol{l}}'^{(i)}\ \cdots]$ and $M' = HM = [\cdots\ \tilde{\boldsymbol{m}}'^{(j)}\ \cdots]$. Find the best-fitting homography H.

We can approach this problem in a step-by-step manner, implying each additional constraint at each step:

- Firstly, to restore affinity, that is, to apply a "pure" homography

$$H_P = \begin{bmatrix} 1 & & \\ & 1 & \\ v_x & v_y & 1 \end{bmatrix}$$

  such that if $L' \overset{H_P}{\mapsto} L_P$ and $M \overset{H_P}{\mapsto} M_P$, then each line in L_P is pairwise parallel, and each line in M_P is pairwise parallel. The process is called **affine rectification**.

- Secondly, to restore similarity, that is, to apply a "pure" affinity

$$H_A = \begin{bmatrix} a_{11} & a_{12} & \\ a_{21} & a_{22} & \\ & & 1 \end{bmatrix}$$

  such that if $L_P \overset{H_A}{\mapsto} L_A$ and $M_P \overset{H_P}{\mapsto} M_A$, then the transformed image is only different from the original one by a similarity. This process is called **metric recrification**.

- Finally, to restore the original coordinates of the lines, that is, to apply a similarity

$$H_S = \begin{bmatrix} k\cos\theta & -k\sin\theta & t_x \\ k\sin\theta & k\cos\theta & t_y \\ & & 1 \end{bmatrix}$$

  such that if $L_A \overset{H_S}{\mapsto} L_S$ and $M_A \overset{H_P}{\mapsto} M_S$, then $L_S = L$ and $M_S = M$, that is the original lines are fully recovered.

4.5.1. Affine rectification

The two vanishing points $\tilde{\boldsymbol{p}}_l^{(\infty)}$ and $\tilde{\boldsymbol{p}}_m^{(\infty)}$ for the ℓ-direction and m-direction can be determined by solving the linear system: $L^T\tilde{\boldsymbol{x}} = \tilde{\boldsymbol{0}}$ and $M^T\tilde{\boldsymbol{x}} = \tilde{\boldsymbol{0}}$. This can be done via the SVD. The vanishing line is therefore $\tilde{\boldsymbol{v}}^{(\infty)} = \tilde{\boldsymbol{p}}_l^{(\infty)} \times \tilde{\boldsymbol{p}}_m^{(\infty)}$.

The homography for affine rectification is thus, $H_P = \begin{bmatrix} 1 & & \\ & 1 & \\ v_x^{(\infty)} & v_y^{(\infty)} & 1 \end{bmatrix}$.

4.5.2. Metric rectification

For metric rectification, we either need a pair of vanishing points and a pair of orthogonal lines, or five pairs of orthogonal lines [7, p.57]. It is clear that for our application, the former option is more optimal.

Consider each pair of lines in the affine-rectified space $\left(\tilde{\boldsymbol{l}}_A^{(i)}, \tilde{\boldsymbol{m}}_A^{(j)}\right)$. Note that the first two components of $\tilde{\boldsymbol{l}}^{(i)} = \left(l_1^{(i)}, l_2^{(i)}, l_3^{(i)}\right)$ and $\tilde{\boldsymbol{m}}^{(j)} = \left(m_1^{(j)}, m_2^{(j)}, m_3^{(j)}\right)$ are the slope vector of the respective lines. Because the original lines are orthogonal, $l_1^{(i)}m_1^{(j)} + l_2^{(i)}m_2^{(j)} = 0$, or in matrix form,

$$\tilde{\boldsymbol{l}}^{(i)T} C_\infty^* \tilde{\boldsymbol{m}}^{(j)} = 0$$

where $C_\infty^* = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 0 \end{bmatrix}$ is the dual conic[3].

An affinity H_A

$$\begin{aligned} &\left(H_A\tilde{\boldsymbol{l}}_A^{(i)}\right)^T C_\infty^* \left(H_A\tilde{\boldsymbol{m}}_A^{(j)}\right) = 0 \\ \Rightarrow\ & \tilde{\boldsymbol{l}}_A^{(i)T}\left(H_A^T C_\infty^* H_A\right)\tilde{\boldsymbol{m}}_A^{(j)} = 0 \\ \Rightarrow\ & \tilde{\boldsymbol{l}}^{(i)T}\begin{bmatrix} \boldsymbol{K}^T & \\ & 1 \end{bmatrix}\begin{bmatrix} \boldsymbol{I} & \\ & 0 \end{bmatrix}\begin{bmatrix} \boldsymbol{K} & \\ & 1 \end{bmatrix}\tilde{\boldsymbol{m}}^{(j)} = 0 \\ \Rightarrow\ & \tilde{\boldsymbol{l}}^{(i)T}\begin{bmatrix} \boldsymbol{K}^T\boldsymbol{K} & \\ & 0 \end{bmatrix}\tilde{\boldsymbol{m}}^{(j)} = 0 \end{aligned}$$

[3] Some remarks about the dual conic

4.5.3. Similarity rectification

4.6. Camera calibration

$$P = KR[I_3 \mid \vec{t}]$$

5. Experiments

5.1. Dataset Preparation

The SoccerNet Calibration Dataset (`sn_calibration`) [4] consists of 12356 training samples, 2796 validation samples and 2719 testing samples. Our actual training and validation processes only use a subset of the given samples for model prototyping purposes.

Task	Training	Validation	Testing
General	12356	2796	2719
Edge segmentation	128	64	64
Keypoint concentration	128	128	128

Table 1:

Each sample contains two files: a `PNG` colour image and a `JSON` data file containing

[4]

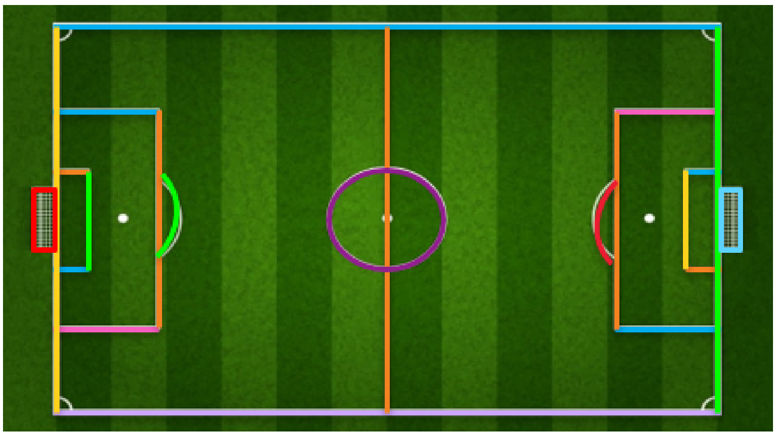


Figure 13: The thing [4]

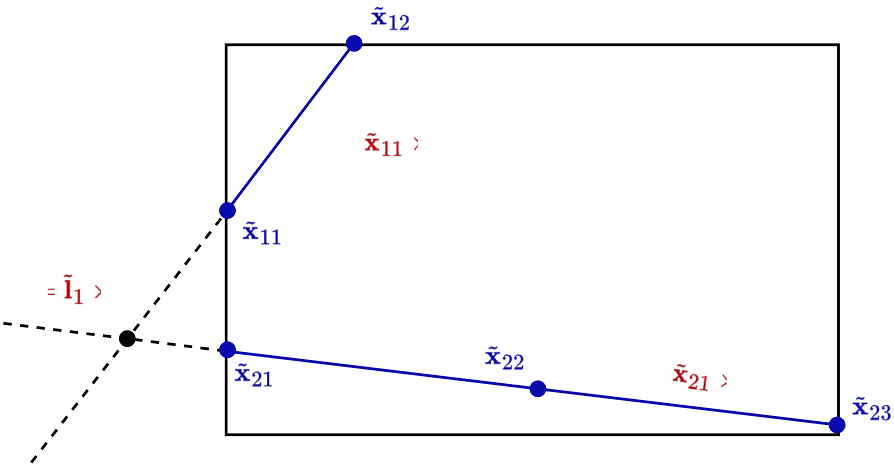


Figure 14: Keypoint refill

For instance, given the coordinates of the field's top and left border lines, the top left corner keypoint can be computed as:

$$\tilde{p}_{\text{FIELD\_TOP\_LEFT}} = \tilde{l}_{\text{FIELD\_BOX\_TOP}} \times \tilde{l}_{\text{FIELD\_BOX\_LEFT}}$$

This can be done for all the visible keypoints on the field, as well as off-screen keypoints whose constituent lines are visible.

Given that at least 4 non-collinear keypoints are positioned, the coordinates of the other of the 27 keypoints can be detected via a homography transformation. This can be done via the DLT algorithm.

5.2. Image segmentation

5.3. Field registration via homography transformation

6. Discussion

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